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Riccati equations for wave propagation in planarly-stratified solids *

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Abstract – Wave propagation and scattering are investigated in three-dimensional, dissipative, anisotropic bodies with one-dimensional inhomogeneity. The constitutive equations are taken to be linear. The incidence is allowed to be oblique. Three different procedures are set up which involve the impedance matrix and two local reflection matrices. All three matrix functions are shown to satisfy appropriate Riccati equations and jump conditions (at discontinuities). The operative aspects are examined to solve reflection–transmission problems for a layer sandwiched between two half spaces. The unknown field within the layer is determined through the impedance or the reflection matrices. © 2000 Éditions scientifiques et médicales Elsevier SAS

wave propagation / stratified solids / reflection and transmission matrices / Riccati equations

1. Introduction

Wave propagation in stratified media is of interest for many applications in seismics, ocean acoustics and non-destructive evaluation. Here we study wave propagation and scattering in three-dimensional, dissipative, anisotropic bodies with one-dimensional inhomogeneity, in the z-direction (planarly-stratified or plane stratified solid). Also, the constitutive equations are taken to be linear. To allow for oblique incidence we let the displacement depend on the transverse coordinates x, y and time t through the factor $\exp[i(k_x x + k_y y - \omega t)]$ where ω is the real frequency and k_x, k_y are possibly-complex wavenumbers in the x, y directions. Hence the equations governing the behaviour of the body turn out to be expressed by the system

$$\mathbf{w}' = \mathbf{A}\mathbf{w}, \tag{1.1}$$

where $\mathbf{w}$ is the set of 6 unknown functions of $z \in (-\infty, \infty)$, $\mathbf{w}' := \mathrm{d}\mathbf{w}/\mathrm{d}z$, and $\mathbf{A}$ is a 6×6 complex-valued matrix function of z. Starting from the system (1.1), the reflection–transmission process results in a boundary-value problem for $\mathbf{w}$. This in turn is expressed as a Cauchy problem for appropriate unknown matrices.

The purpose of this paper is to establish efficient procedures for the determination of the solution to the reflection–transmission problem at a planarly-stratified half space. This means that, given an incident wave in the homogeneous half space $z < 0$, we determine the reflected wave in $z < 0$ and the transmitted one in $z > 0$. In addition to be planarly-stratified, the half space $z \geqslant 0$ is allowed to contain jump discontinuities for the material properties. Three procedures are developed, for three matrices, which provide global properties, such as reflection and transmission of layers and multilayers, without requiring details about the solution $\mathbf{w}$. The results of this paper for the 6×6 system (1.1) may be generalized, under appropriate assumptions, for any $2n \times 2n$ system in the unknown $2n$-tuple $\mathbf{w}$.

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First, partition $\mathbf{w}$ into two 3-component vectors $\mathbf{w}^I, \mathbf{w}^{II}$ such that the column vector $[\mathbf{w}^I, \mathbf{w}^{II}]^T$ is $\mathbf{w}$. Here $\mathbf{w}^I$ is the displacement and $\mathbf{w}^{II}$ is the traction at planes $z =$ constant. We then consider the admittance (or impedance) matrix $\mathbf{Y}$ such that $\mathbf{w}^{II} = \mathbf{Y}\mathbf{w}^I$. Hence we determine a differential equation for $\mathbf{Y}(z)$ such that, once an appropriate boundary condition is given, the function $\mathbf{Y}$ is established in terms of the material properties of the body.

Secondly, let $\mathbf{p}_\alpha$, $\alpha = 1, \ldots, 6$, be the eigenvectors of $\mathbf{A}$ with eigenvalues d_α. We express $\mathbf{w}$ as

$$\mathbf{w} = \sum_{\alpha=1}^{6} s_\alpha \mathbf{p}_\alpha.$$

Again, we partition the 6-tuple $s_1, \ldots, s_6$ into two 3-component vectors $\mathbf{s}^f, \mathbf{s}^b$ and consider Σ such that $\mathbf{s}^b = \Sigma \mathbf{s}^f$. Hence we determine a differential equation for $\Sigma(z)$. Once an appropriate boundary condition is given, the function Σ is established in terms of the material properties of the body. The eigenvector $\mathbf{p}_\alpha$ is the α-th polarization vector. In homogeneous bodies the components of $\mathbf{s}(z)$ are proportional to $\exp(i d_\alpha z)$. Along with the factor $\exp(-i\omega t)$ they are interpreted as forward- or backward- propagating waves according to the value of d_α. Incidentally, $d_1, \ldots, d_6$ need not be pairwise complex conjugate; they are so in elastic materials. We merely assume that, e.g., $s_1\mathbf{p}_1, \ldots, s_3\mathbf{p}_3$ are forward-propagating and $s_4\mathbf{p}_4, \ldots, s_6\mathbf{p}_6$ backward-propagating.

Thirdly, technical reasons suggest that we consider a different unknown, $\mathbf{v}$, related to $\mathbf{s}$ by $\mathbf{s} = \mathbf{E}\mathbf{v}$ where $\mathbf{E}$ is a diagonal matrix involving the exponentials of integrals of the eigenvalues of $\mathbf{A}$. The entries of $\mathbf{E}$ take the form $\exp(i \int_0^z d_\alpha(\xi)\, \mathrm{d}\xi)$. In homogeneous bodies each component of $\mathbf{v}$ is the amplitude of a plane wave $\exp[i(d_\alpha z - \omega t)]$. Still we partition $\mathbf{v}$ into forward- and backward-propagating parts, $\mathbf{v}^f$ and $\mathbf{v}^b$, and introduce the local reflection matrix $\mathbf{R}$ such that $\mathbf{v}^b = \mathbf{R}\mathbf{v}^f$. Meanwhile we investigate the local transmission matrix $\mathbf{T}$.

All matrix functions $\mathbf{Y}, \Sigma, \mathbf{R}$ are shown to satisfy appropriate Riccati equations. The linear differential equation for $\mathbf{T}$ is also derived. The matrix $\mathbf{Y}$ is continuous even at points where $\mathbf{A}$ is discontinuous. Meanwhile, Σ and $\mathbf{R}$ are discontinuous and the related jump conditions are obtained. Once the functions $\mathbf{Y}$, Σ or $\mathbf{R}$ are found, the field $\mathbf{w}$ at any point z is determined. For definiteness, specific results are established for a layer in $z \in (0, d)$ sandwiched between two homogeneous half spaces. Also, a numerical investigation is performed of the influence of the thickness on the reflection matrix of a layer.

A reference to previous works and known procedures on related subjects is in order. The solution to (1.1) is investigated through the propagator matrix Ω such that $\Omega' = \mathbf{A}\Omega$, $\Omega(0) = \mathbf{1}$ (cf. (Kennett (1984)). The lack of an explicit expression for Ω is the main drawback of the method. In Thompson–Haskell method (Thomson, 1950; Haskell, 1953) the propagator matrix is expressed by appropriate exponentials. Caviglia and Morro (1992) generalize the method to dissipative layers. The drawback, however, is that the representation holds in homogeneous bodies only. Brekhovskikh (1980) investigates continuously layered media through an equation of the form $\psi'' + K^2(z)\psi = 0$. The approach is confined to one-dimensional (scalar) problems.

Lately, some improvements in wave propagation in layered media have been realized. Within the time domain, Romeo (1999) investigates wave propagation in homogeneous solids. Corones and Karlsson (1988) restrict attention to isotropic solids (Ammicht et al. (1987) consider only normal incidence) and show that reflection and transmission are given by convolutions involving appropriate kernels. Both kernels are found to satisfy integro-differential equations which involve partial derivatives of the unknown function.

More often approaches are developed in the frequency domain where (1.1) is the governing equation. Ursin (1984) follows an approach which may be viewed as a particular case of the description via $\mathbf{s}$ and Σ for $n = 1$. Kennett (1984) restricts attention to isotropic solids and finds that the reflection matrix of the layer (our $\mathbf{R}(0_+)$) in terms of d satisfies a Riccati equation to be solved with the condition $\mathbf{R}(0_+) = 0$ as $d = 0$. Tromp and Snieder (1989) apply to P- and S-waves an approach similar to that of Kennett (1984). Lewicki et al. (1996) determine

the Riccati equation for $\mathbf{R}(z)$ and let $\mathbf{R}(d) = 0$. Romeo (1996) finds that, on the assumption of normal incidence and of convenient symmetries for the anisotropic tensor, the reflectivity obeys a Riccati equation.

The use of the impedance matrix is less frequent. We mention Hager and Rostamian (1988) for elastic and isotropic solids. Titchener and Willis (1991) use the impedance for electromagnetic solids and argue on $\mathbf{Y}(z)$ by the relation for $\mathbf{Y}(0)$.

None of these papers allows for jump discontinuities at the boundary of the layer. Nor do they look for the unknown field $\mathbf{w}$ in the layer $z \in (0, d)$. The approaches of the present paper allow for any planar inhomogeneity and any number of discontinuity surfaces. They provide the reflection and transmission matrices of the whole body (multilayer) through a Riccati equation together with an initial value which accounts for the pertinent boundary condition.

2. First-order system for time-harmonic waves

We consider a linear, viscoelastic, anisotropic stratified solid. The vector function $\mathbf{u}(\mathbf{x}, t)$ denotes the displacement, at time t, of the particle occupying the position $\mathbf{x}$ in the equilibrium unstressed configuration. We represent $\mathbf{x}$ by Cartesian coordinates x, y, z such that the material parameters depend on z only. Let $\mathbf{e}_z$ be the unit vector of the z-axis. To fix ideas we view the z-axis as horizontal, the x-axis as vertical.

Body-force effects are neglected and hence the equation of motion reads

$$\rho\ddot{\mathbf{u}} = \nabla \cdot \mathcal{T},$$

where ρ is the density and $\mathcal{T}$ is the Cauchy stress tensor. We look for time-harmonic waves of the form

$$\mathbf{u}(\mathbf{x}, t) = \hat{\mathbf{u}}(z) \exp\big[i(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\big],$$

where ω is the real frequency and $\mathbf{k}_\parallel$ is a given complex-valued wave vector, perpendicular to $\mathbf{e}_z$. Hence $\hat{\mathbf{u}}$ is the unknown vector function of z.

Current models of viscoelasticity result in a linear relation

$$\mathcal{T} = \mathbf{C}\partial\mathbf{u}/\partial\mathbf{x} \qquad (\mathcal{T}_{jh} = C_{jhkl}\partial u_k/\partial x_l),$$

where $\mathbf{C}$ is a complex-valued fourth-order tensor, parameterized by ω, endowed with the standard symmetry properties. As a function of z, $\mathbf{C}$ is taken to be piecewise continuous thus allowing for discontinuity surfaces between layers.

To obtain a first-order system of equations it is convenient to consider the traction at planes perpendicular to the z-axis,

$$\mathbf{t} = \mathcal{T}\mathbf{e}_z = \hat{\mathbf{t}} \exp\big[i(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\big].$$

Upon the observation that

$$\nabla = \frac{\partial}{\partial \mathbf{x}_\parallel} + \mathbf{e}_z \frac{\partial}{\partial z},$$

where $\mathbf{x}_\parallel$ is the part of $\mathbf{x}$ perpendicular to $\mathbf{e}_z$, substitution in the equation of motion and some rearrangements yield (1.1) where $\mathbf{w} = [\hat{\mathbf{u}}, \hat{\mathbf{t}}]^T$. Hereafter we find it convenient to represent a 6×6 matrix through four 3×3 blocks and use the superscripts $I, II, \ldots, IV$, in the form

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}^{I} & \mathbf{A}^{II} \\ \mathbf{A}^{III} & \mathbf{A}^{IV} \end{bmatrix}.$$

In particular, we have

$$\mathbf{A}^{I} = -i(\mathbf{e}_z\mathbf{C}\mathbf{e}_z)^{-1}(\mathbf{e}_z\mathbf{C}\mathbf{k}_\parallel), \qquad \mathbf{A}^{II} = (\mathbf{e}_z\mathbf{C}\mathbf{e}_z)^{-1},$$
$$\mathbf{A}^{III} = -\rho\omega^2\mathbf{1} + (\mathbf{k}_\parallel\mathbf{C}\mathbf{k}_\parallel) - (\mathbf{k}_\parallel\mathbf{C}\mathbf{e}_z)(\mathbf{e}_z\mathbf{C}\mathbf{e}_z)^{-1}(\mathbf{e}_z\mathbf{C}\mathbf{k}_\parallel), \qquad \mathbf{A}^{IV} = \big(\mathbf{A}^{I}\big)^{T}.$$

For any two vectors **a** and **b** here we let **aCb** be a second-order tensor defined by

$$(\mathbf{aCb})_{jk} = a_h C_{hjkl} b_l.$$

As shown by Caviglia and Morro (1999), the thermodynamic requirement $\Im\mathbf{C} < 0$ implies that $\mathbf{e}_z\mathbf{C}\mathbf{e}_z$ is non-singular and then that **A** exists. Hence the dynamics of a solid is described by a first-order system of the form (1.1) where **w** is the column vector $[\hat{\mathbf{u}}, \hat{\mathbf{t}}]^T$, namely the traction-displacement vector.

3. Impedance matrix

The system (1.1) with $\mathbf{w} = [\hat{\mathbf{u}}, \hat{\mathbf{t}}]^T \in \mathbb{C}^6$ can be written as

$$\hat{\mathbf{u}}' = \mathbf{A}^{I}\hat{\mathbf{u}} + \mathbf{A}^{II}\hat{\mathbf{t}}, \qquad \hat{\mathbf{t}}' = \mathbf{A}^{III}\hat{\mathbf{u}} + \mathbf{A}^{IV}\hat{\mathbf{t}}. \tag{3.1}$$

Let **Y** be the (local) impedance matrix defined by

$$\hat{\mathbf{t}} = \mathbf{Y}\hat{\mathbf{u}}.$$

Differentiation with respect to z and comparison with the expression of $\hat{\mathbf{t}}'$ in (3.1) yields

$$\big(\mathbf{Y}' + \mathbf{Y}\mathbf{A}^{I} + \mathbf{Y}\mathbf{A}^{II}\mathbf{Y} - \mathbf{A}^{III} - \mathbf{A}^{IV}\mathbf{Y}\big)\hat{\mathbf{u}} = 0.$$

The arbitrariness of $\hat{\mathbf{u}}$ implies that the impedance matrix satisfies the Riccati differential equation

$$\mathbf{Y}' = \mathbf{A}^{III} + \mathbf{A}^{IV}\mathbf{Y} - \mathbf{Y}\mathbf{A}^{I} - \mathbf{Y}\mathbf{A}^{II}\mathbf{Y}. \tag{3.2}$$

If $\mathbf{Y}(z_0)$ is known for some z_0 then the integration, in the existence domain $I \ni z_0$, provides $\mathbf{Y}(z)$, $z \in I$. Once **Y** is determined, the function $\hat{\mathbf{u}}$ is found by solving the linear differential equation

$$\hat{\mathbf{u}}' = \big(\mathbf{A}^{I} + \mathbf{A}^{II}\mathbf{Y}\big)\hat{\mathbf{u}}$$

with a suitable value $\hat{\mathbf{u}}(z_0)$. The traction $\hat{\mathbf{t}}$ is then determined by applying **Y** to $\hat{\mathbf{u}}$.

Some remarks are in order about two particular cases. First, let $z = z_0$ be a free boundary whence $\hat{\mathbf{u}}(z_0)$ is undetermined while $\hat{\mathbf{t}}(z_0) = 0$ and $\mathbf{Y}(z_0) = 0$. Hence **Y** is determined by integration but $\hat{\mathbf{u}}$ is determined only if $\hat{\mathbf{u}}(\bar{z})$ is known at some other point $\bar{z} \neq z_0$. Secondly, let $z = z_0$ be a fixed boundary whence $\hat{\mathbf{u}}(z_0) = 0$ while $\hat{\mathbf{t}}(z_0)$ is undetermined and $\mathbf{Y}(z_0)$ is regarded as undetermined or infinite. In such a case it is convenient to consider the inverse matrix $\mathbf{Y}^{-1} = \mathbf{K}$, which then satisfies the Riccati differential equation

$$\mathbf{K}' = \mathbf{A}^{I}\mathbf{K} + \mathbf{A}^{II} - \mathbf{K}\mathbf{A}^{IV} - \mathbf{K}\mathbf{A}^{III}\mathbf{K},$$

and let $\mathbf{K}(z_0) = 0$. Once $\mathbf{K}$ is determined then $\hat{\mathbf{u}} = \mathbf{K}\hat{\mathbf{t}}$ provides $\hat{\mathbf{u}}$ in terms of $\hat{\mathbf{t}}$. Since $\hat{\mathbf{t}}(z_0)$ is undetermined, it is required that $\hat{\mathbf{t}}(\bar{z})$ is known at some other point $\bar{z} \neq z_0$.

By way of example, consider an isotropic solid where

$$C_{ijkl}(z) = \lambda(z)\delta_{ij}\delta_{kl} + \mu(z)(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

If the solid is viscoelastic then the Lamé coefficients λ and μ are given by

$$\lambda = \lambda_0 + \int_0^\infty \lambda'(\eta)\exp(i\omega\eta)\,\mathrm{d}\eta, \qquad \mu = \mu_0 + \int_0^\infty \mu'(\eta)\exp(i\omega\eta)\,\mathrm{d}\eta,$$

where $\lambda_0, \mu_0 \in \mathbb{R}$ are taken to be positive and $\lambda', \mu' \in L^1(\mathbb{R})$ are real-valued. The dependence of $\lambda_0, \mu_0, \lambda', \mu'$ on z is understood. The requirement $\Im\mathbf{C} < 0$ amounts to $\Im\mu < 0$, $\Im(2\mu + 3\lambda) < 0$. Moreover let $\mathbf{k}_\parallel$ have a real direction; we choose the x-axis in the direction of $\mathbf{k}_\parallel$ and denote by k_x the component of $\mathbf{k}_\parallel$. The system (1.1) then decouples into two subsystems,

$$\mathbf{w}_v' = \mathbf{A}_v\mathbf{w}_v, \qquad \mathbf{w}_h' = \mathbf{A}_h\mathbf{w}_h, \tag{3.3}$$

where

$$\mathbf{w}_v = \begin{bmatrix} \hat{u}_x \\ \hat{u}_z \\ \hat{t}_x \\ \hat{t}_z \end{bmatrix}, \qquad \mathbf{A}_v = \begin{bmatrix} 0 & -ik_x & 1/\mu & 0 \\ -i\gamma k_x & 0 & 0 & \gamma/\lambda \\ \zeta k_x^2 - \rho\omega^2 & 0 & 0 & -i\gamma k_x \\ 0 & -\rho\omega^2 & -ik_x & 0 \end{bmatrix},$$

$$\mathbf{w}_h = \begin{bmatrix} \hat{u}_y \\ \hat{t}_y \end{bmatrix}, \qquad \mathbf{A}_h = \begin{bmatrix} 0 & 1/\mu \\ -\mu\sigma_T^2 & 0 \end{bmatrix},$$

where

$$\gamma = \lambda/(\lambda + 2\mu), \qquad \zeta = \mu(\lambda + \mu)(\lambda + 2\mu), \qquad \sigma_T = \sqrt{\rho\omega^2/\mu - k_x^2}.$$

Since $\rho\omega^2/\mu - k_x^2 \in \mathbb{C}$ and $\Im(\rho\omega^2/\mu) > 0$, for small values of $\Im k_x^2$ we have $\Im(\rho\omega^2/\mu - k_x^2) > 0$. Henceforth we let the square root σ_T belong to the first quadrant.

The variables $\mathbf{w}_v$ and $\mathbf{w}_h$ describe vertically- and horizontally-polarized waves. The decoupling (3.3) means that the two polarizations are conserved throughout the body.

Let $Y_h \in \mathbb{C}$ be the impedance, associated with $\mathbf{w}_h$, such that $\hat{t}_y = Y_h\hat{u}_y$. By the form of the 2×2 matrix $\mathbf{A}_h$ we have $A^I, A^{IV} = 0$, $A^{II} = 1/\mu$, $A^{III} = -\mu\sigma_T^2$. Hence (3.2) becomes

$$Y_h' = \mu k_x^2 - \rho\omega^2 - \frac{1}{\mu}Y_h^2. \tag{3.4}$$

4. Wave modes

Consider again equation (1.1). If the matrix $\mathbf{A}$ is constant and simple (cf. Lancaster and Tismenetsky (1985)), the general integral of (1.1) is expressed as a linear combination of suitable exponential functions times the constant eigenvectors of $\mathbf{A}$. We assume that $\mathbf{A}$ is simple at each value of z and we follow a similar procedure which is made operative by a change of the variables describing the wave field.

Denote by id_α the (not necessarily distinct) eigenvalues of $\mathbf{A}$ and by $\mathbf{p}_\alpha = [\mathbf{u}_\alpha, \mathbf{t}_\alpha]^T$ the related independent eigenvectors. Here α varies from 1–6. We assume that the eigenvalues are non-zero. We introduce the matrices $\mathbf{P}(z)$, whose ordered columns are $\mathbf{p}_1, \ldots, \mathbf{p}_6$, and $\mathbf{D}(z)$, which is defined by

$$\mathbf{D} = \operatorname{diag}[d_1, \ldots, d_6].$$

It is a well known result that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = i\mathbf{D}. \tag{4.1}$$

Next define the new dependent variables $\mathbf{s}(z)$ through the equation

$$\mathbf{w} = \mathbf{P}\,\mathbf{s}, \tag{4.2}$$

where $\mathbf{s}$ is an unknown 6-dimensional column vector. The differential equations for $\mathbf{s}$ are found by substitution of (4.2) in (1.1) and comparison with (4.1). It follows that

$$\mathbf{s}' = i\mathbf{D}\,\mathbf{s} - \mathbf{Q}\,\mathbf{s}, \tag{4.3}$$

where $\mathbf{Q} = \mathbf{P}^{-1}\,\mathbf{P}'$.

Once $\mathbf{s}(z)$ is determined, by (4.2) we obtain the unknown $[\mathbf{u}, \mathbf{t}]^T$ in the form

$$\begin{bmatrix}\mathbf{u}\\ \mathbf{t}\end{bmatrix} = \mathbf{w}\exp\big[i(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\big] = \sum_{\alpha=1}^{6} s_\alpha(z)\,\mathbf{p}_\alpha(z)\exp\big[i(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\big]. \tag{4.4}$$

In the particular case of uniform media, $\mathbf{A}$, $\mathbf{D}$ and $\mathbf{P}$ are constant matrices and hence

$$\mathbf{s}(z) = \operatorname{diag}\big[\exp(id_1 z), \ldots, \exp(id_6 z)\big]\,\mathbf{c},$$

where $\mathbf{c}$ is any constant column vector. Substitution in (4.4) gives

$$\begin{bmatrix}\mathbf{u}\\ \mathbf{t}\end{bmatrix} = \sum_{\alpha=1}^{6} c_\alpha \mathbf{p}_\alpha \exp(i\mathbf{k}_\parallel \cdot \mathbf{x})\exp\big[i(d_\alpha z - \omega t)\big],$$

which allows $[\mathbf{u}, \mathbf{t}]^T$ to be viewed as a linear combination of inhomogeneous waves Caviglia and Morro (1992). That is why the components s_α, though in the general case of stratified media, are viewed as representatives of wave modes or eigenmodes (cf. Lewicki et al. (1996)).

The off-diagonal terms in (4.3) account for the phenomenon of mode conversion in the course of propagation; because of mode conversion, excitation of a single wave mode inside a layer is not allowed under general conditions.

As usual, we assume that the six modes are partitioned into three forward-propagating and three backward-propagating. For definiteness we let $\mathbf{s} = [\mathbf{s}^f, \mathbf{s}^b]^T$ where $\mathbf{s}^f = [s_1, s_2, s_3]^T$ and $\mathbf{s}^b = [s_4, s_5, s_6]^T$ represent forward- and backward-propagating modes. Let Σ be the matrix such that

$$\mathbf{s}^b = \Sigma\,\mathbf{s}^f. \tag{4.5}$$

It is convenient to write (4.3) in the block-form

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{bmatrix}\mathbf{s}^f\\ \mathbf{s}^b\end{bmatrix} = i\begin{bmatrix}\mathbf{D}^I & \mathbf{0}\\ \mathbf{0} & \mathbf{D}^{IV}\end{bmatrix}\begin{bmatrix}\mathbf{s}^f\\ \mathbf{s}^b\end{bmatrix} - \begin{bmatrix}\mathbf{Q}^I & \mathbf{Q}^{II}\\ \mathbf{Q}^{III} & \mathbf{Q}^{IV}\end{bmatrix}\begin{bmatrix}\mathbf{s}^f\\ \mathbf{s}^b\end{bmatrix}.$$

Hence, differentiation of (4.5) and comparison provide the differential equation for Σ in the form

$$\Sigma' = i\left(\mathbf{D}^{IV}\Sigma - \Sigma\mathbf{D}^{I}\right) - \mathbf{Q}^{III} + \Sigma\mathbf{Q}^{I} - \mathbf{Q}^{IV}\Sigma + \Sigma\mathbf{Q}^{II}\Sigma. \tag{4.6}$$

The connection between $\mathbf{Y}$ and Σ is determined by observing that, by (4.2),

$$\begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{Y}\hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{P}^{I} & \mathbf{P}^{II} \\ \mathbf{p}^{III} & \mathbf{P}^{IV} \end{bmatrix} \begin{bmatrix} \mathbf{s}^{f} \\ \Sigma\mathbf{s}^{b} \end{bmatrix},$$

the dependence on z being understood. Evaluation of $\hat{\mathbf{u}}$ and substitution yields

$$\mathbf{Y} = \left(\mathbf{P}^{III} + \mathbf{P}^{IV}\Sigma\right)\left(\mathbf{P}^{I} + \mathbf{P}^{II}\Sigma\right)^{-1}. \tag{4.7}$$

It is worth looking at (4.6) in the particular case of horizontally-polarized waves $\mathbf{w}_h$ in (3.3). The pertinent matrices are given by

$$\mathbf{D}_h = \begin{bmatrix} \sigma_T & 0 \\ 0 & -\sigma_T \end{bmatrix}, \qquad \mathbf{P}_h = \begin{bmatrix} 1/\mu & 1/\mu \\ i\sigma_T & -i\sigma_T \end{bmatrix},$$

and

$$\hat{u}_y = \frac{1}{\mu}\left(s^{f} + s^{b}\right), \qquad \hat{t}_y = i\sigma_T\left(s^{f} - s^{b}\right).$$

Also,

$$Q^{I} = Q^{IV} = -\frac{1}{2}\left(\frac{\mu'}{\mu} - \frac{\sigma_T'}{\sigma_T}\right), \qquad Q^{II} = Q^{III} = -\frac{1}{2}\left(\frac{\mu'}{\mu} + \frac{\sigma_T'}{\sigma_T}\right).$$

Hence (4.6) gives

$$\Sigma_h' = -2i\sigma_T\Sigma_h + \frac{1}{2}\left(\frac{\mu'}{\mu} + \frac{\sigma_T'}{\sigma_T}\right)\left(1 - \Sigma_h^2\right). \tag{4.8}$$

Meanwhile, by (4.7) we obtain

$$Y_h = i\mu\sigma_T\frac{1 - \Sigma_h}{1 + \Sigma_h}.$$

A similar procedure yields the differential equation for the 4×4 matrix Σ_v.

5. Wave amplitudes

The occurrence of $\mathbf{s}' - i\mathbf{D}\mathbf{s}$ suggests that the evolution equation simplifies by multiplying (4.3) by $\exp[-i\int_0^z \mathbf{D}(\tau)\,\mathrm{d}\tau]$. We then have

$$\frac{\mathrm{d}}{\mathrm{d}z}\left\{\exp\left[-i\int_0^z \mathbf{D}(\tau)\,\mathrm{d}\tau\right]\mathbf{s}\right\} = \exp\left[-i\int_0^z \mathbf{D}(\xi)\,\mathrm{d}\xi\right]\mathbf{Q}\mathbf{s}.$$

Hence, letting

$$\mathbf{E}(z) = \exp\left[i\int_0^z \mathbf{D}(\xi)\,\mathrm{d}\xi\right]$$

and

$$\mathbf{v} = \mathbf{E}^{-1}\mathbf{s}$$

we can write (4.3) as

$$\mathbf{v}' = -\mathbf{E}^{-1}\mathbf{Q}\mathbf{E}\mathbf{v}. \tag{5.1}$$

The meaning of $\mathbf{v}$ follows by observing that, by definition,

$$\mathbf{w} = \mathbf{P}\mathbf{E}\mathbf{v} \tag{5.2}$$

and hence

$$\mathbf{w}\exp\left[i\,(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\right] = \sum_{\alpha=1}^{6} v_\alpha(z)\exp\left[i\int_0^z d_\alpha(\tau)\,\mathrm{d}\tau\right]\mathbf{p}_\alpha(z)\exp\left[i\,(\mathbf{k}_\parallel \cdot \mathbf{x} - \omega t)\right].$$

This ascribes to the components v_α of $\mathbf{v}$ the meaning of local amplitudes of the propagation modes. In the case of uniform media d_α and $\mathbf{p}_\alpha$ are constant, $\mathbf{Q} = 0$, $\mathbf{v}$ is constant, and the modes are expressed by inhomogeneous waves.

As with $\mathbf{s}$, we let the first three components v_1, v_2, v_3 describe forward-propagating waves while v_4, v_5, v_6 describe backward-propagating waves. Let $\mathbf{v}^f = (v_1, v_2, v_3)$ and $\mathbf{v}^b = (v_4, v_5, v_6)$. The local reflection matrix $\mathbf{R}$ is defined to map $\mathbf{v}^f$ into $\mathbf{v}^b$ at any value of z. Again it is convenient to consider the block form of the evolution equation. Let $\mathbf{F} = -\mathbf{E}^{-1}\mathbf{Q}\mathbf{E}$ so that (5.1) takes the form

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{bmatrix}\mathbf{v}^f\\ \mathbf{v}^b\end{bmatrix} = \begin{bmatrix}\mathbf{F}^I & \mathbf{F}^{II}\\ \mathbf{F}^{III} & \mathbf{F}^{IV}\end{bmatrix}\begin{bmatrix}\mathbf{v}^f\\ \mathbf{v}^b\end{bmatrix}. \tag{5.3}$$

Hence, differentiation of $\mathbf{v}^b = \mathbf{R}\mathbf{v}^f$ and comparison give

$$\mathbf{R}' = \mathbf{F}^{III} + \mathbf{F}^{IV}\mathbf{R} - \mathbf{R}\mathbf{F}^I - \mathbf{R}\mathbf{F}^{II}\mathbf{R}. \tag{5.4}$$

The connection of $\mathbf{R}$ with Σ and $\mathbf{Y}$ is now established. Since $\mathbf{s} = \mathbf{E}\mathbf{v}$ and $\mathbf{E}$ is diagonal, applying the argument for (4.7) we obtain

$$\Sigma = \mathbf{E}^{IV}\mathbf{R}\left(\mathbf{E}^I\right)^{-1}. \tag{5.5}$$

Substitution for Σ from (5.5) in (4.7) yields

$$\mathbf{Y} = \left[\mathbf{P}^{III} + \mathbf{P}^{IV}\mathbf{E}^{IV}\mathbf{R}\left(\mathbf{E}^I\right)^{-1}\right]\left[\mathbf{P}^I + \mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{R}\left(\mathbf{E}^I\right)^{-1}\right]^{-1}. \tag{5.6}$$

The inverse relation is

$$\mathbf{R} = \left(\mathbf{E}^{IV}\right)^{-1}\left(\mathbf{Y}\mathbf{P}^{II} - \mathbf{P}^{IV}\right)^{-1}\left(\mathbf{P}^{III} - \mathbf{Y}\mathbf{P}^I\right)\mathbf{E}^I. \tag{5.7}$$

A simple, interesting, example is provided by the horizontally-polarized waves. In such a case the pertinent $\mathbf{v}$ is a two-component vector, say $[v^f, v^b]$, where

$$v^f = \frac{1}{2}\left(\frac{\mu}{E_T^-}\hat{u}_y + \frac{1}{i\sigma_T E_T^-}\hat{t}_y\right), \qquad v^b = \frac{1}{2}\left(\frac{\mu}{E_T^+}\hat{u}_y - \frac{1}{i\sigma_T E_T^+}\hat{t}_y\right)$$

and

$$E_T^{\pm}(z) = \exp\left[\pm i \int_0^z \sigma_T(\tau)\,\mathrm{d}\tau\right].$$

The components v^f, v^b satisfy the system

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{bmatrix} v^f \\ v^b \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \mu'/\mu - \sigma_T'/\sigma_T & (\mu'/\mu + \sigma_T'/\sigma_T)(E_T^-)^2 \\ (\mu'/\mu + \sigma_T'/\sigma_T)(E_T^+)^2 & \mu'/\mu - \sigma_T'/\sigma_T \end{bmatrix}\begin{bmatrix} v^f \\ v^b \end{bmatrix}.$$

Both $R_h = v^b/v^f$ and $F^I, \ldots, F^{IV}$ are scalar quantities. Moreover $F^{IV}R_h - R_h F^I$ vanishes. Hence R is found to satisfy the Riccati equation (5.4) in the simple form

$$R_h' = \frac{1}{2}\left(\frac{\mu'}{\mu} + \frac{\sigma_T'}{\sigma_T}\right)(E_T^+)^2\left[1 - (E_T^-/E_T^+)^2 R_h^2\right]. \tag{5.8}$$

Let $z = l$ be the point where transmitted waves are considered (end of a layer or of a multilayer). We define the transmission matrix $\mathbf{T}(z)$ such that

$$\mathbf{T}(z)\mathbf{v}^f(z) = \mathbf{v}^f(l), \qquad \mathbf{T}(l) = \mathbf{1}.$$

If a discontinuity for $\mathbf{A}$ occurs at $z = l$ then $\mathbf{T}(l)$ and $\mathbf{v}^f(l)$ stand for the values at $z = l_-$ or $z = l_+$ accordingly as the transmitted wave is considered at l_- or l_+. Differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\mathbf{T}(z)\mathbf{v}^f(z)\right] = 0.$$

Substitution for $\mathrm{d}\mathbf{v}^f/\mathrm{d}z$ from (5.3) yields

$$\mathbf{T}' = -\mathbf{T}\mathbf{F}^I - \mathbf{T}\mathbf{F}^{II}\mathbf{R}. \tag{5.9}$$

Equation (5.9) provides $\mathbf{T}(z)$, as $z < l$, once $\mathbf{R}(z)$ is known. For horizontally-polarized waves $\mathbf{T}$ reduces to a scalar function, T_h, and (5.9) simplifies to

$$T_h' = -\left(\frac{\mu'}{\mu} + \frac{\sigma_T'}{\sigma_T}\right)(E_T^-)^2 R_h T_h. \tag{5.10}$$

6. Jumps of R and T at a discontinuity

Let $z = \bar{z}$ be a discontinuity surface for $\mathbf{A}$. This means that the entries of $\mathbf{A}$ are taken to be C^1 functions in $(\bar{z} - \delta, \bar{z}) \cup (\bar{z}, \bar{z} + \delta)$ for some $\delta > 0$, and to suffer a jump discontinuity across $z = \bar{z}$.

As a surface force, $\mathbf{t}$ is continuous across any surface. If a surface realizes a welded contact then also $\mathbf{u}$ is continuous. Hence, by the assumption of welded contact, the function $\mathbf{w}$ is continuous across $z = \bar{z}$, even though $\mathbf{A}$ is not. As a consequence $\mathbf{Y}$ is continuous across $z = \bar{z}$. Look now at $\mathbf{R}$ and Σ. Since $\mathbf{w} = \mathbf{P}\mathbf{E}\mathbf{v}$ then

$$\mathbf{P}(z_-)\mathbf{E}(z)\mathbf{v}(z_-) = \mathbf{P}(z_+)\mathbf{E}(z)\mathbf{v}(z_+)$$

holds at any value of z. Hence we have

$$\mathbf{v}(z_-) = \mathbf{H}(z)\mathbf{v}(z_+), \tag{6.1}$$

where

$$\mathbf{H}(z)=\mathbf{E}^{-1}(z)\mathbf{P}^{-1}(z_-)\mathbf{P}(z_+)\mathbf{E}(z).$$

Equation (6.1) provides the jump of $\mathbf{v}$.

To derive the relation between $\mathbf{R}(z_-)$ and $\mathbf{R}(z_+)$ we write (6.1) as

$$\begin{bmatrix}\mathbf{v}^f(z_-)\\ \mathbf{R}(z_-)\mathbf{v}^f(z_-)\end{bmatrix}=\begin{bmatrix}\mathbf{H}^I(z) & \mathbf{H}^{II}(z)\\ \mathbf{H}^{III}(z) & \mathbf{H}^{IV}(z)\end{bmatrix}\begin{bmatrix}\mathbf{v}^f(z_+)\\ \mathbf{R}(z_+)\mathbf{v}^f(z_+)\end{bmatrix}.$$

Evaluation of $\mathbf{v}^f(z_-)$ and substitution allow us to find that

$$\mathbf{R}(z_-)\big[\mathbf{H}^I(z)+\mathbf{H}^{II}(z)\mathbf{R}(z_+)\big]\mathbf{v}^f(z_+)=\big[\mathbf{H}^{III}(z)+\mathbf{H}^{IV}(z)\mathbf{R}(z_+)\big]\mathbf{v}^f(z_+).$$

The arbitrariness of $\mathbf{v}^f(z_+)$ allows us to conclude that

$$\mathbf{R}(z_-)=\big[\mathbf{H}^{III}(z)+\mathbf{H}^{IV}(z)\mathbf{R}(z_+)\big]\big[\mathbf{H}^I(z)+\mathbf{H}^{II}(z)\mathbf{R}(z_+)\big]^{-1} \tag{6.2}$$

provided only that the indicated inverse matrix exists. The jump $\mathbf{R}(z_+)\mapsto\mathbf{R}(z_-)$ is determined by (6.2) as soon as the material properties (and then $\mathbf{H}$) around the discontinuity (at z_-, z_+) are known.

In practical problems we are asked to establish an appropriate initial value of $\mathbf{R}$. Three significant cases are examined. First, if $z>l$ is a homogeneous half-space and no wave is incoming then only outgoing waves may occur. Hence $\mathbf{v}^f(l_+)\neq 0$ while $\mathbf{v}^b(l_+)=0$ and we set $\mathbf{R}(l_+)=0$. Hence $\mathbf{R}(l_-)$, the initial value for (5.4), is given by (6.2).

The other two cases involve free boundaries and fixed boundaries. In this regard observe that

$$\mathbf{w}=\begin{bmatrix}\hat{\mathbf{u}}\\ \hat{\mathbf{t}}\end{bmatrix}=\begin{bmatrix}\mathbf{P}^I\mathbf{E}^I & \mathbf{P}^{II}\mathbf{E}^{IV}\\ \mathbf{P}^{III}\mathbf{E}^I & \mathbf{P}^{IV}\mathbf{E}^{IV}\end{bmatrix}\begin{bmatrix}\mathbf{v}^f\\ \mathbf{v}^b\end{bmatrix}. \tag{6.3}$$

Hence the condition of zero displacement and zero traction at $z=l$ become

$$\mathbf{u}=0\Longleftrightarrow\mathbf{P}^I\mathbf{E}^I\mathbf{v}^f(z_-)+\mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{v}^b(z_-)=0, \tag{6.4}$$

$$\mathbf{t}=0\Longleftrightarrow\mathbf{P}^{III}\mathbf{E}^I\mathbf{v}^f(z_-)+\mathbf{P}^{IV}\mathbf{E}^{IV}\mathbf{v}^b(z_-)=0. \tag{6.5}$$

In both cases we obtain $\mathbf{R}(z_-)$.

The jump for $\mathbf{T}$ is determined as follows. The definition of $\mathbf{T}(z)$ and the jump condition (6.1) for $\mathbf{v}$ give

$$\mathbf{T}(z_-)\mathbf{v}^f(z_-)=\mathbf{T}(z_+)\mathbf{v}^f(z_+).$$

Replacing $\mathbf{v}^f(z_-)$ with $\mathbf{H}^I(z)+\mathbf{H}^{II}(z)\mathbf{R}(z_+)$ we obtain

$$\mathbf{T}(z_+)=\mathbf{T}(z_-)\big[\mathbf{H}^I(z)+\mathbf{H}^{II}(z)\mathbf{R}(z_+)\big]$$

providing the jump $\mathbf{T}(z_-)\mapsto\mathbf{T}(z_+)$.

7. Direction of propagation and energy flux

There are different definitions of wave direction which are shown to be non-equivalent, Caviglia and Morro (1999). A characterization of the direction is then in order and we think that the appropriate one is to say that a wave is going in a direction if the energy flux in that direction is positive.

As shown in the previous sections, we need to select forward- and backward-propagating waves. We now show how this is operative through the energy flux. The time-averaged energy flux vector **J** of a time-harmonic mechanical wave is given by (cf. Boulanger and Hayes (1993); Caviglia and Morro (1992))

$$\mathbf{J} = \frac{1}{2}\omega\Im\{\mathbf{T}\mathbf{u}^*\}.$$

The component of **J** along the direction $\mathbf{e}_z$, say $\mathcal{F}$, is defined by

$$\mathcal{F} = \mathbf{J}\cdot\mathbf{e}_z = \frac{1}{2}\omega\Im\{\mathbf{t}\cdot\mathbf{u}^*\}.$$

Hence, we find that

$$\mathcal{F} = \mathbf{w}^*\cdot\mathcal{I}\mathbf{w},$$

where

$$\mathcal{I} = \frac{i\omega}{4}\begin{bmatrix}\mathbf{0} & -\mathbf{1}\\ \mathbf{1} & \mathbf{0}\end{bmatrix},$$

where **0** and **1** are the null matrix and the identity matrix in $\mathbb{R}^3$. Substitution of the representation (4.4) yields

$$\mathcal{F} = \sum_{\alpha,\beta=1}^{6} s_\alpha^*\Phi_{\alpha\beta}s_\beta\exp\bigl[-2\Im(\mathbf{k}_\parallel\cdot\mathbf{x})\bigr],$$

where

$$\Phi_{\alpha\beta} = \mathbf{p}_\alpha^\dagger\mathcal{I}\mathbf{p}_\beta$$

and a superposed dagger means transpose and complex conjugate. Owing to the definition of $\mathcal{I}$ it follows that the matrix Φ is Hermitian, $\Phi = \Phi^\dagger$. Accordingly, the diagonal entries $\Phi_{\alpha\alpha}$ are real.

The diagonal entries $\Phi_{\alpha\alpha}$ enter the component $\mathcal{F}$ through $s_\alpha^*\Phi_{\alpha\alpha}s_\alpha\exp[-2\Im(\mathbf{k}_\parallel\cdot\mathbf{x})]$ which denotes the local mean energy flux of the α-th mode. For the single α-th mode we have

$$\operatorname{sgn}\mathcal{F} = \operatorname{sgn}\Phi_{\alpha\alpha}$$

and hence $\mathbf{p}_\alpha$ represents a forward- or backward-propagating wave according as $\Phi_{\alpha\alpha} > 0$ or $\Phi_{\alpha\alpha} < 0$. The off-diagonal entries of Φ provide cross terms among the pertinent modes.

In viscoelastic stratified solids the sign of $\Phi_{\alpha\alpha}$ is not necessarily constant along propagation. This in turn means that the partition in three forward- and three backward-propagating waves may cease to hold, thus making the wave splitting inapplicable. We show in a moment that, for horizontally-polarized waves, the partition works provided k_x is real.

Consider horizontally-polarized waves. Upon substitution of the corresponding eigenvectors, say $\mathbf{p}_1, \mathbf{p}_2$, we obtain the 2×2 submatrix Φ in the form

$$\Phi = \frac{1}{2}\omega\begin{bmatrix}\Re(\sigma_T/\mu^*) & -i\Im(\sigma_T/\mu^*)\\ i\Im(\sigma_T/\mu^*) & -\Re(\sigma_T/\mu^*)\end{bmatrix}.$$

Hence we have

$$\mathcal{F}(z) = \frac{1}{2}\omega\bigl[\Re(\sigma_T/\mu^*)\bigl(|s^f|^2 - |s^b|^2\bigr) + 2\,\Im(\sigma_T/\mu^*)\Im\bigl(s^b s^{f*}\bigr)\bigr]\exp\bigl[-2\Im(\mathbf{k}_\parallel\cdot\mathbf{x})\bigr].$$

Indeed, the dependence of $\mathcal{F}$ on z is through the quantity in brackets, namely through the material properties σ_T, μ and the coefficients s^f, s^b which are governed by the appropriate subsystem of (4.3).

Since σ_T is in the first quadrant, if, further, $\Re\mu > 0$ as is the case in elasticity, then we find that

$$\Re\left(\sigma_T/\mu^*\right) > 0.$$

Meanwhile the sign of $\Im(\sigma_T/\mu^*)$ is left undetermined.

8. Reflection and transmission by a layer

We now show how the classical problem of reflection–transmission by a layer can be investigated. Since we have in mind a planarly-inhomogeneous layer in $(0, d)$, we let the material parameters ρ, $\mathbf{C}$ be constant in $(-\infty, 0) \cup (d, \infty)$ and continuous in $(0, d)$. Jump discontinuities of ρ and $\mathbf{C}$ at $z = 0, d$ are allowed to occur. We require that $\mathbf{w}$ is continuous everywhere and hence also at $z = 0, d$.

To fix ideas, the incident wave is coming from $-\infty$ and hence the reflected wave propagates back in the half space $z < 0$ while the transmitted wave propagates forward in the half space $z > d$. Owing to causality, the reflected and transmitted waves are superpositions of backward- and forward-propagating modes, respectively.

The procedure is based on the Riccati equation (5.4) for $\mathbf{R}(z)$ and the linear equation (5.9) for $\mathbf{T}(z)$, the jumps of $\mathbf{R}$, $\mathbf{T}$ at a discontinuity, and the appropriate boundary conditions, e.g. (6.5). The incident, reflected, and transmitted waves are represented by $\mathbf{v}^i = \mathbf{v}^f(0_-)$, $\mathbf{v}^r = \mathbf{v}^b(0_-)$, $\mathbf{v}^t = \mathbf{v}^f(d_+)$. The reflection matrix of the layer is $\mathbf{R}(0_-)$ in that

$$\mathbf{v}^r = \mathbf{R}(0_-)\mathbf{v}^i.$$

By (6.2) we can write $\mathbf{R}(0_-)$ in terms of $\mathbf{R}(0_+)$,

$$\mathbf{R}(0_-) = \left[\mathbf{H}^{III}(0) + \mathbf{H}^{IV}(0)\mathbf{R}(0_+)\right]\left[\mathbf{H}^{I}(0) + \mathbf{H}^{II}(0)\mathbf{R}(0_+)\right]^{-1}.$$

The value of $\mathbf{R}(0_+)$ is determined by integrating (5.4) once $\mathbf{R}(d_-)$ is given. In turn, the value $\mathbf{R}(d_-)$ is related to the half space $z > d$. Examine two cases. First, the half-space $z > d$ is empty. Hence $\mathbf{t}$ vanishes at $z = d_\pm$ and, by (6.5), we have

$$\mathbf{R}(d_-) = -\left[\left(\mathbf{P}^{IV}\mathbf{E}^{IV}\right)^{-1}\mathbf{P}^{III}\mathbf{E}^{I}\right](d_-).$$

Secondly, the half space $z > d$ is homogeneous and no wave comes from $+\infty$. Hence, $\mathbf{v}^b(d_+) = 0$ and $\mathbf{R}(d_+) = 0$. By (6.2) we have

$$\mathbf{R}(d_-) = \mathbf{H}^{III}(d)(\mathbf{H}^{I})^{-1}(d).$$

The transmission matrix of the layer is $\mathbf{T}(0_-)$ in that

$$\mathbf{v}^t = \mathbf{T}(0_-)\mathbf{v}^i.$$

To determine $\mathbf{T}(0_-)$ we integrate (5.9) subject to $\mathbf{T}(d_+) = \mathbf{1}$, if the half space $z > d$ is non-empty, or $\mathbf{T}(d_-) = \mathbf{1}$, if $z > d$ is empty.

As a by-product, once the functions $\mathbf{R}(z)$ and $\mathbf{T}(z)$ are determined, we can evaluate $\mathbf{v}^f(z)$ and $\mathbf{v}^b(z)$ in the layer $z \in (0, d)$. For, by $\mathbf{T}(0_-)\mathbf{v}^i = \mathbf{T}(z)\,\mathbf{v}^f(z)$ we have

$$\mathbf{v}^f(z) = \mathbf{T}^{-1}(z)\,\mathbf{T}(0_-)\,\mathbf{v}^i.$$

Moreover,

$$\mathbf{v}^b(z) = \mathbf{R}(z)\,\mathbf{v}^f(z).$$

An alternative procedure allows the reflection matrix of the layer, $\mathbf{R}(0_-)$, to be determined through the impedance $\mathbf{Y}$. Given $\mathbf{Y}(d)$, possibly through (5.6) in terms of $\mathbf{R}(d_+)$ or $\mathbf{R}(d_-)$, the integration of (3.2) yields $\mathbf{Y}(0)$ and hence, by (5.7),

$$\mathbf{R}(0_-) = \big[\mathbf{Y}(0)\mathbf{P}^{II}(0_-) - \mathbf{P}^{IV}(0_-)\big]^{-1}\big[\mathbf{P}^{III}(0_-) - \mathbf{Y}(0)\mathbf{P}^{I}(0_-)\big]. \tag{8.1}$$

Owing to the Riccati equations for $\mathbf{R}$ or $\mathbf{Y}$, in general definite results can be obtained by numerical integration. In this regard equation (3.2) contains the values of the material functions through $\mathbf{A}$ but does not contain the derivatives which, instead, occur in equation (5.4) for $\mathbf{R}$. In favour of $\mathbf{Y}$ we mention that equation (5.4) for $\mathbf{R}$ and equation (4.6) for Σ involve the eigenvectors $\mathbf{p}_\alpha$ and their derivatives $\mathbf{p}'_\alpha$ through $\mathbf{F}$; their evaluation is usually cumbersome, which motivates numerical (approximate) approaches. Also, $\mathbf{Y}$ is more convenient in representing $\mathbf{w}$. Meanwhile $\mathbf{R}$ gives direct information on the amplitudes of the waves and is essential in the characterization of the boundary conditions.

Sometimes the reflection is described through backward- and forward-parts of $\hat{\mathbf{u}}$ or $\hat{\mathbf{t}}$ (cf. Achenbach (1975)). This view is now generalized by considering the corresponding matrices $\mathbf{R}_\mathbf{u}$ and $\mathbf{R}_\mathbf{t}$. Consider equation (6.3) and define the forward- and backward-parts of $\hat{\mathbf{u}}$, $\hat{\mathbf{t}}$ by

$$\hat{\mathbf{u}}^f = \mathbf{P}^I\mathbf{E}^I\mathbf{v}^f, \qquad \hat{\mathbf{u}}^b = \mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{v}^b,$$
$$\hat{\mathbf{t}}^f = \mathbf{P}^{III}\mathbf{E}^I\mathbf{v}^f, \qquad \hat{\mathbf{t}}^b = \mathbf{P}^{IV}\mathbf{E}^I\mathbf{v}^b.$$

Hence

$$\hat{\mathbf{u}}^b = \mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{R}\mathbf{v}^f = \mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{R}\big(\mathbf{E}^I\big)^{-1}\big(\mathbf{P}^I\big)^{-1}\hat{\mathbf{u}}^f.$$

Quite naturally we define $\mathbf{R}_\mathbf{u}$ by

$$\hat{\mathbf{u}}^b = \mathbf{R}_\mathbf{u}\hat{\mathbf{u}}^f.$$

Comparison gives

$$\mathbf{R}_\mathbf{u} = \mathbf{P}^{II}\mathbf{E}^{IV}\mathbf{R}\big(\mathbf{E}^I\big)^{-1}\big(\mathbf{P}^I\big)^{-1}. \tag{8.2}$$

By the same token we find that

$$\mathbf{R}_\mathbf{t} = \mathbf{P}^{IV}\mathbf{E}^{IV}\mathbf{R}\big(\mathbf{E}^I\big)^{-1}\big(\mathbf{P}^{III}\big)^{-1}.$$

Also, letting $\mathbf{T}_\mathbf{u}$ be such that $\mathbf{T}_\mathbf{u}(z)\hat{\mathbf{u}}^f(z) = \hat{\mathbf{u}}^f(l)$ we find that

$$\mathbf{T}_\mathbf{u}(z) = \mathbf{E}^I(l)\mathbf{P}^I(l)\mathbf{T}(z)\big(\mathbf{E}^I\big)^{-1}(z)\big(\mathbf{P}^I\big)^{-1}(z). \tag{8.3}$$

By replacing $\mathbf{P}^I$ with $\mathbf{P}^{III}$ we obtain the relation for $\mathbf{T}_\mathbf{t}$.

The effect of dissipation due to viscoelasticity is involved because the complex-valued terms of $\mathbf{A}$ enter almost at any step. However, a remarkable effect occur for $\mathbf{R}_\mathbf{u}$ and $\mathbf{T}_\mathbf{u}$, and similarly for $\mathbf{R}_\mathbf{t}$ and $\mathbf{T}_\mathbf{t}$. Look at (8.2) and (8.3). The (diagonal) entries of $(\mathbf{E}^I)^{-1}$ and of $\mathbf{E}^{IV}$ take, e.g., the form $\exp(-i\int_0^z \Re\sigma_T(\xi)\,\mathrm{d}\xi)\exp(-\int_0^z \Im\sigma_T(\xi)\,\mathrm{d}\xi$. Since $\Im\sigma_T > 0$ it follows that, in passing from the wave modes $\mathbf{v}$ to the components of $\mathbf{u}$, reflection is enhanced. On the contrary, $\mathbf{T}_\mathbf{u}(0)$ is affected by $\mathbf{E}^I(l)$ and hence, in passing from the wave modes $\mathbf{v}$ to the components of $\mathbf{u}$, transmission is contrasted. Qualitatively, this shows that dissipation favours reflection and hinders transmission.

9. Applications

In addition to providing a general procedure for a numerical investigation of wave propagation, the Riccati equation allows us to find the (exact) solution in particular cases. This is exemplified as follows.

9.1. Horizontally-polarized waves in isotropic bodies

The 2×2 matrices $\mathbf{D}_h, \mathbf{P}_h, \mathbf{E}_h$ are involved. Also let $\phi = \mu\sigma_T$. The reflection coefficients $R_{\mathbf{u}h}$ and $R_{\mathbf{t}h}$ for displacement and traction are related to R_h by

$$R_{\mathbf{u}h}(z) = \left(E_T^-\right)^2(z)R_h(z), \qquad R_{\mathbf{t}h}(z) = -\left(E_T^-\right)^2(z)R_h(z).$$

The incident wave propagates in the homogeneous half-space $(-\infty, 0)$. The half space $z > 0$ consists of a homogeneous layer in $(0, d)$ and a homogeneous half space in (d, ∞). It is understood that $\phi(0_-) \neq \phi(0_+)$. We evaluate the reflection coefficient $\mathcal{R} := R_h(0_-)$ in two cases: (1) Two-layer medium, in that the half space $z \in (d, \infty)$ is empty; (2) Three-layer medium, in that the half space $z \in (d, \infty)$ is homogeneous and $\phi(d_-) \neq \phi(d_+)$. The homogeneity of the layers makes R_h more convenient than Y_h.

Two-layer medium. To fix $R_h(d_-)$ we observe that, by (6.5),

$$i\phi(d_-)\exp(i\sigma_T d)v^f(d_-) - i\phi(d_-)\exp(-i\sigma_T d)v^b(d_-) = 0$$

whence

$$R_h(d_-) = \frac{v^b}{v^f}(d_-) = \exp(2i\sigma_T d).$$

By (5.8) it follows that $R_h' \equiv 0$ in $(0, d)$ and hence $R_h(0_+) = R_h(d_-)$. Now,

$$R_h(0_-) = \frac{H_h^{III}(0) + H_h^{IV}(0)R_h(0_+)}{H_h^{I}(0) + H_h^{II}(0)R_h(0_+)}$$

and

$$\mathbf{H}_h(0) = \mathbf{P}_h^{-1}(0_-)\mathbf{P}(0_+) = \frac{1}{2\mu(0_+)\sigma_T(0_-)}\begin{bmatrix}\phi(0_-)+\phi(0_+) & \phi(0_-)-\phi(0_+)\\ \phi(0_-)-\phi(0_+) & \phi(0_-)+\phi(0_+)\end{bmatrix}.$$

Hence we obtain

$$R_h(0_-) = \frac{1 + \kappa\frac{\exp(id\sigma_T)-\exp(-id\sigma_T)}{\exp(id\sigma_T)+\exp(-id\sigma_T)}}{1 - \kappa\frac{\exp(id\sigma_T)-\exp(-id\sigma_T)}{\exp(id\sigma_T)+\exp(-id\sigma_T)}}, \tag{9.1}$$

where $\kappa = \phi(0_+)/\phi(0_-)$. If σ_T is real then

$$R_h(0_-) = \frac{1 + \kappa\tan(d\sigma_T)}{1 - \kappa\tan(d\sigma_T)}. \tag{9.2}$$

This equation is often termed the Fresnel formula.

If, instead, σ_T is complex-valued, let $\sigma_T = \alpha + i\beta$, where $\alpha, \beta > 0$. As $\beta d \gg 1$ we have

$$R_h(0_-) \simeq \frac{1 - \kappa\exp(-2i\alpha d)}{1 + \kappa\exp(-2i\alpha d)}.$$

Accordingly, as $\beta d \gg 1$, $R_h(0_-)$ is a periodic, complex-valued, function of the thickness d.

Three-layer medium. At d_+ only the forward-propagating wave occurs, $v^b(d_+) = 0$, and hence $R_h(d_+) = 0$. By (6.2) we have

$$R_h(d_-) = \frac{H_h^{III}(d)}{H_h^{I}(d)}.$$

By the definition of $\mathbf{H}(z)$ we find that

$$\mathbf{H}_h(d) = \frac{\mu(d_-)}{2\mu(d_+)} \begin{bmatrix} 1+\phi(d_+)/\phi(d_-) & [1-\phi(d_+)/\phi(d_-)]\exp(-2id\sigma_T) \\ [1-\phi(d_+)/\phi(d_-)]\exp(2id\sigma_T) & 1+\phi(d_+)/\phi(d_-) \end{bmatrix}.$$

Hence we have

$$R_h(d_-) = \frac{[\phi(d_-) - \phi(d_+)]\exp(2id\sigma_T)}{\phi(d_-) + \phi(d_+)}.$$

By the homogeneity of the layer $(0, d)$, equation (5.8) provides

$$R_h(0_+) = R_h(d_-).$$

Also, by (6.2), upon evaluation of $\mathbf{H}_h$ at $z = 0$ we have

$$R_h(0_-) = \frac{\phi(0_-) - \phi(0_+) + [\phi(0_-) + \phi(0_+)]R_h(0_+)}{\phi(0_-) + \phi(0_+) + [\phi(0_-) - \phi(0_+)]R_h(0_+)}$$

whence

$$R_h(0_-) = \frac{[\phi(0_-) - \phi(0_+)][\phi(d_-) + \phi(d_+)] + [\phi(0_-) + \phi(0_+)][\phi(d_-) - \phi(d_+)]\exp(2id\sigma_T)}{[\phi(0_-) + \phi(0_+)][\phi(d_-) + \phi(d_+)] + [\phi(0_-) - \phi(0_+)][\phi(d_-) - \phi(d_+)]\exp(2id\sigma_T)}.$$

Also as a check of consistency, we examine the limit case where the thickness d approaches zero. In the two-layer medium, $\tan(d\sigma_T) \to 0$ gives

$$R_h(0_-) \to 1 \tag{9.3}$$

which, as expected, is the result for a empty half space. In the three-layer medium, subject to $\phi(d_-) = \phi(d_+)$, we have

$$R_h(0_-) \to \frac{\phi(0_-) - \phi(0_+)}{\phi(0_-) + \phi(0_+)}. \tag{9.4}$$

This limit is the reflection coefficient associated with a discontinuity between two homogeneous half spaces. Indeed, in connection with (9.4) we find that

$$R_{\mathbf{u}h} \to \frac{\phi(0_-) - \phi(0_+)}{\phi(0_-) + \phi(0_+)}, \qquad R_{\mathbf{t}h} \to \frac{\phi(0_+) - \phi(0_-)}{\phi(0_-) + \phi(0_+)}.$$

If μ is real (no dissipation) and the incidence is normal then σ_T is the wavenumber and $\phi = \mu\omega/v = \rho\omega v$, where v is the wave speed. The standard relation occurring in the literature (cf. Achenbach (1975))

$$R = \frac{\rho_+ v_+ - \rho_- v_-}{\rho_+ v_+ + \rho_- v_-}$$

is then recovered as a particular case.

Numerical results are now determined for the following parameters:

$$\rho = 8.9\ \mathrm{g/cm^3}, \qquad \omega = 10^5\ \mathrm{Hz}, \qquad k_x = 0.093\ \mathrm{cm^{-1}},$$

and

$$\mu = 44.44\,10^{10}\ \mathrm{g/cm\,sec^2}, \qquad z < 0,$$

$$\mu = (1 + 2z/d)4.444 \times 10^{11} - i(z/d)2.086 \times 10^5\ \mathrm{g/cm\,sec^2}, \qquad z \in (0, d),$$

$$\mu = 3 \times 4.444 \times 10^{11} - i2.086 \times 10^5\ \mathrm{g/cm\,sec^2}, \qquad z > d.$$

The half space $z < 0$ is regarded as elastic where μ takes the value for (the elastic model of) annealed copper and $z > d$ is a viscoelastic half space where $\Re\mu$ is three times the value for the elastic half space and $\Im\mu$ is the value for the dissipative model of annealed copper. The factor 3 is chosen so as to make more evident the effects of the inhomogeneity of the layer. For simplicity, the function $\mu(z)$, as $s \in (0, d)$, is taken to be linear.

Figures 1 and *2* represent the real and imaginary parts R_h versus z in a slab of thickness $d = 1$ cm. The results are obtained by integrating (3.4) to determine $Y_h(z)$ and then by passing to $R_h(z)$ through (5.7).

Figures 3 and *4* represent the real and imaginary parts of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer. As $d \to 0$ we have $R_h(0_-) \to -0.246$ while the imaginary part is negligible, $\simeq 10^{-7}$. Meanwhile, by (9.4) we obtain the same conclusion. The results are obtained by integrating (3.4) and then applying (8.1).

Next a two-layer medium is considered ($z > d$ is empty). *Figures 5* and *6* represent the real and imaginary parts of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer. It is worth remarking that the inhomogeneity of the layer rules out the periodic character of $R_h(0_-)$ as a function of d, cf. (9.1). As $d \to 0$ we have $R_h(0_-) \to 1$, which agrees with (9.3).

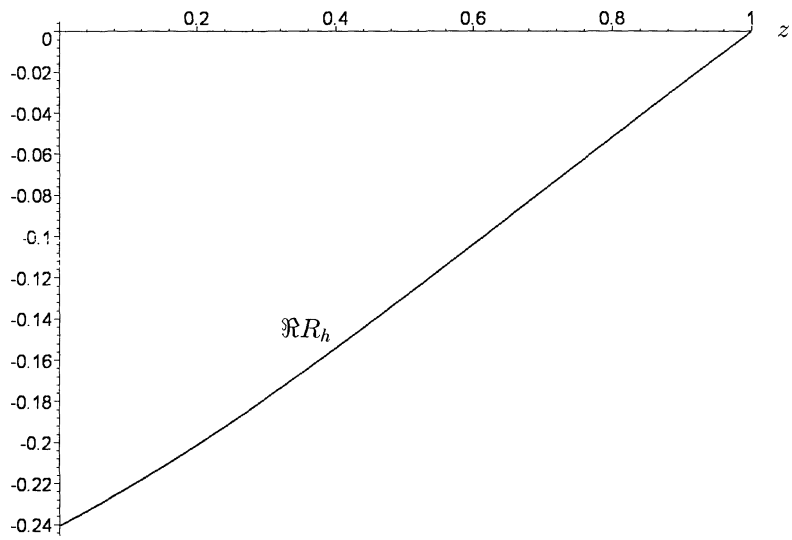


Figure 1. The real part of R_h versus z in a layer of thickness $d = 1$ cm.

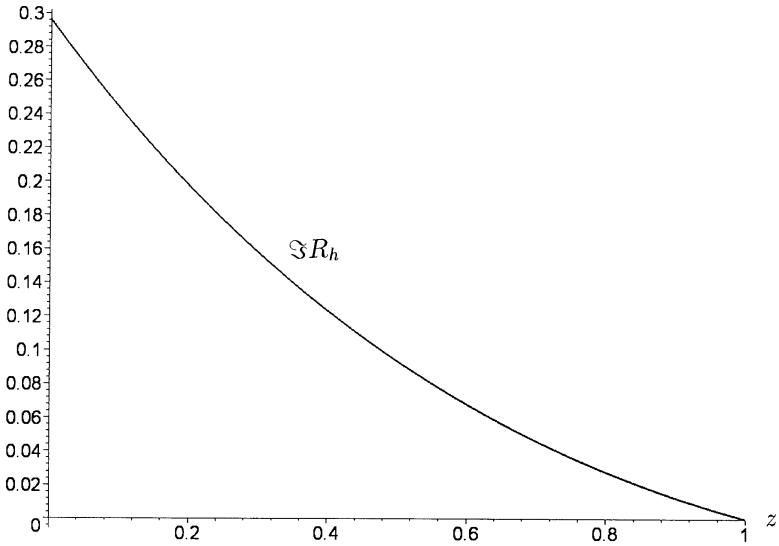


Figure 2. The imaginary part of R_h versus z in a layer of thickness $d = 1$ cm.

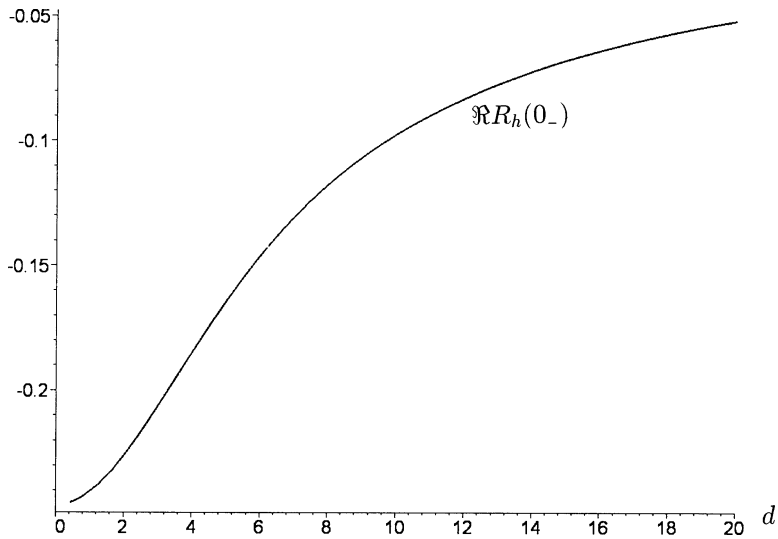


Figure 3. The real part of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer in a three-layer medium.

9.2. Vertically-polarized waves in isotropic bodies

Preliminarily we look at normal incidence, namely to $k_x = 0$. Hence

$$\mathbf{A}_v = \begin{bmatrix} 0 & 0 & 1/\mu & 0 \\ 0 & 0 & 0 & \gamma/\lambda \\ -\rho\omega^2 & 0 & 0 & 0 \\ 0 & -\rho\omega^2 & 0 & 0 \end{bmatrix}.$$

Accordingly, the system $\mathbf{w}_v' = \mathbf{A}_v \mathbf{w}_v$ decouples into two subsystems,

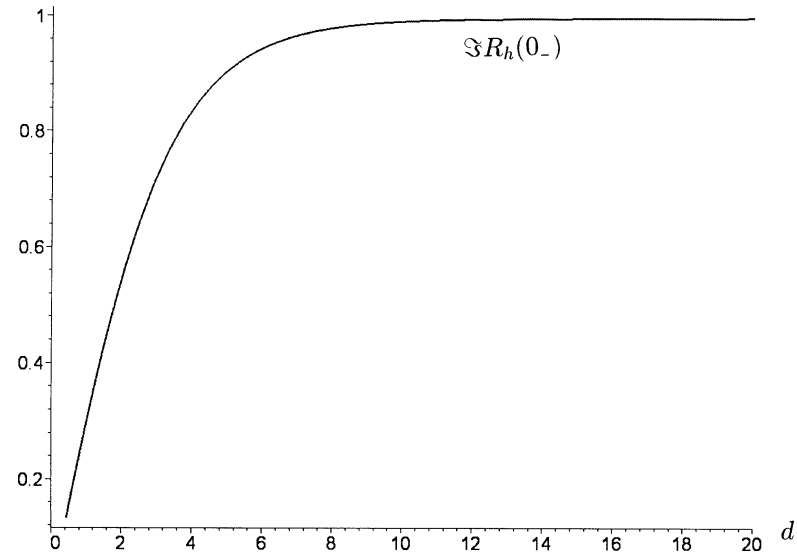


Figure 4. The imaginary part of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer in a three-layer medium.

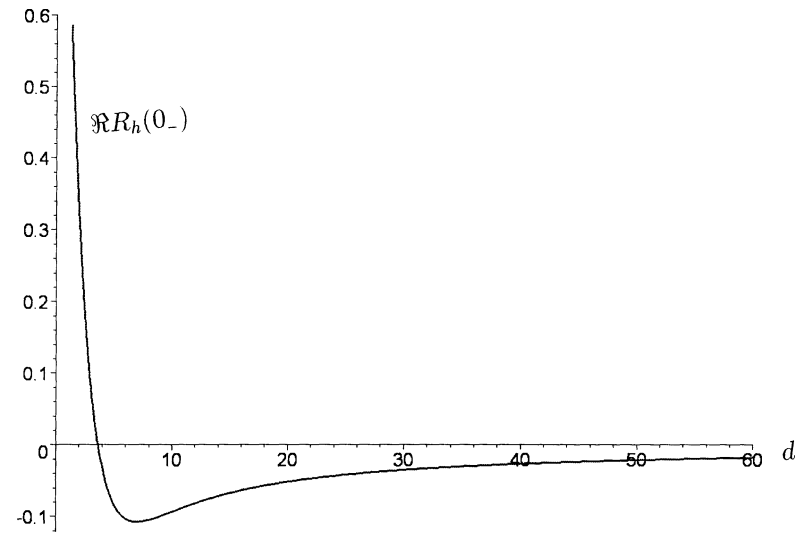


Figure 5. The real part of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer in a two-layer medium.

$$\begin{bmatrix} u_x \\ t_x \end{bmatrix}' = \begin{bmatrix} 0 & 1/\mu \\ -\rho\omega^2 & 0 \end{bmatrix} \begin{bmatrix} u_x \\ t_x \end{bmatrix}, \qquad \begin{bmatrix} u_z \\ t_z \end{bmatrix}' = \begin{bmatrix} 0 & \gamma/\lambda \\ -\rho\omega^2 & 0 \end{bmatrix} \begin{bmatrix} u_z \\ t_z \end{bmatrix}.$$

This means that transverse (x) and longitudinal (z) polarizations are conserved as the wave propagates in planarly-stratified solids (in the z-direction). Also, the continuity of **u** and **t** at interfaces makes the polarization invariant under reflection and trasmission processes. Hence, upon purely formal changes, reflection and transmission due to normal incidence parallel the behaviour of horizontally-polarized waves.

Let $k_x \neq 0$. Longitudinal and transverse polarizations are now coupled. Also,

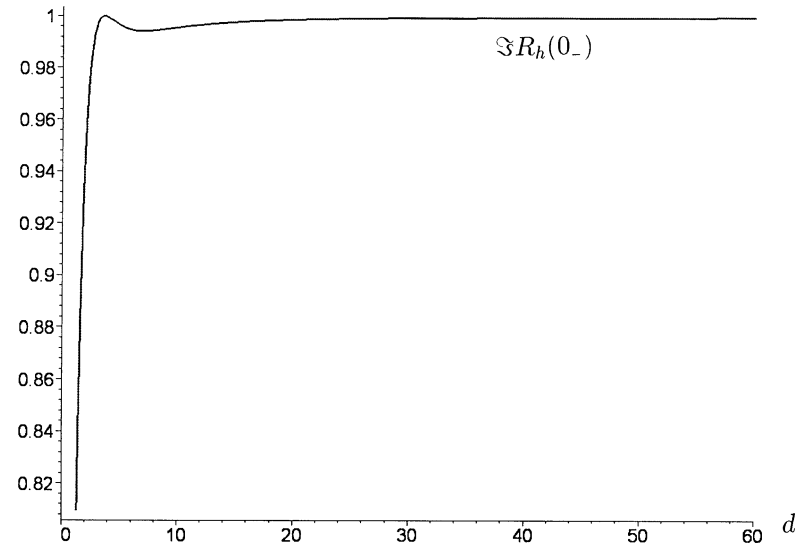


Figure 6. The imaginary part of the reflection coefficient $R_h(0_-)$ versus the thickness d of the layer in a two-layer medium.

$$\mathbf{P}_v = \begin{bmatrix} k_x & -\sigma_T & k_x & \sigma_T \\ \sigma_L & k_x & -\sigma_L & k_x \\ 2i\mu\sigma_L k_x & -b & -2i\mu\sigma_L k_x & -b \\ b & 2i\mu\sigma_T k_x & b & -2i\mu\sigma_T k_x \end{bmatrix},$$

where $b = i(\rho\omega^2 - 2\mu k_x^2)$ and

$$\mathbf{D}_v = \operatorname{diag}[i\sigma_L, i\sigma_T, -i\sigma_L, -i\sigma_T],$$

where $\sigma_L = [\rho\omega^2/(\lambda+2\mu) - k_x^2]^{1/2}$, $\Re\sigma_L > 0$, $\Im\sigma_L > 0$. Hence we have

$$\mathbf{P}_v^{-1} = \frac{1}{2i\rho\omega^2} \begin{bmatrix} 2i\mu k_x & b/\sigma_L & k_x/\sigma_L & 1 \\ -b/\sigma_T & 2i\mu k_x & -1 & k_x/\sigma_T \\ 2i\mu k_x & -b/\sigma_L & -k_x/\sigma_L & 1 \\ b/\sigma_T & 2i\mu k_x & -1 & -k_x/\sigma_T \end{bmatrix}.$$

Meanwhile,

$$\mathbf{Q}_v = \mathbf{P}_v^{-1}\mathbf{P}_v' = \frac{1}{2i\rho\omega^2} \begin{bmatrix} b'+a_L & p\sigma_T - b'k_x/\sigma_L & b'-a_L & -p\sigma_T - b'k_x/\sigma_L \\ -p\sigma_L + b'k_x/\sigma_T & b'+a_T & p\sigma_L + b'k_x/\sigma_T & b'-a_T \\ b'-a_L & p\sigma_T + b'k_x/\sigma_L & b'+a_L & -p\sigma_T + b'k_x/\sigma_L \\ -p\sigma_L - b'k_x/\sigma_T & b'-a_T & p\sigma_L - b'k_x/\sigma_T & b'+a_T \end{bmatrix},$$

where

$$a_L = \frac{b\sigma_L + 2ik_x^2(\mu\sigma_L)'}{\sigma_L}, \qquad a_T = \frac{b\sigma_T + 2ik_x^2(\mu\sigma_T)'}{\sigma_T}, \qquad p = 2ik_x\mu', \qquad q = i\rho\omega^2\frac{(\mu\sigma_T)'}{\mu\sigma_T}.$$

Denote by $\mathbf{Q}_v^I, \ldots, \mathbf{Q}_v^{IV}$ the four 2×2 blocks of $\mathbf{Q}_v$ taken in the usual order. Hence equation (5.4) applies where $\mathbf{F}^I, \ldots, \mathbf{F}^{IV}$ are given by

$$\mathbf{F}_v^I = -\mathbf{E}_v^- \mathbf{Q}_v^I \mathbf{E}_v^+, \qquad \mathbf{F}_v^{II} = -\mathbf{E}_v^- \mathbf{Q}_v^{II} \mathbf{E}_v^-, \qquad \mathbf{F}_v^{III} = -\mathbf{E}_v^+ \mathbf{Q}_v^{III} \mathbf{E}_v^+, \qquad \mathbf{F}_v^{IV} = -\mathbf{E}_v^+ \mathbf{Q}_v^{II} \mathbf{E}_v^-,$$

where

$$\mathbf{E}_v^{\pm}(z) = \operatorname{diag}\left[\exp\left(\pm i\int_0^z \sigma_L(\xi)\right)\mathrm{d}\xi,\ \exp\left(\pm i\int_0^z \sigma_T(\xi)\,\mathrm{d}\xi\right)\right].$$

Hence, the reflection matrix $\mathbf{R}_v(0_-)$ for an incident vertically-polarized wave is evaluated upon integration of the appropriate form of (5.4) for $\mathbf{R}_v$, where $\mathbf{F}$ is replaced with $\mathbf{F}_v$, once the initial value for $\mathbf{R}_v$ is known. This problem is exemplified in the case of a layer in $(0, d)$ followed by an empty half space in (d, ∞).

Since $\mathbf{t} = 0$ at $z = d$, by (6.5) we have

$$\mathbf{R}_v(d_-) = -\left[\mathbf{E}_v^+\left(\mathbf{P}_v^{IV}\right)^{-1}\mathbf{P}_v^{III}\mathbf{E}_v^-\right](d_-).$$

Upon integration of (5.4) we then determine $\mathbf{R}_v(0_+)$. In view of the possible discontinuity at $z = 0$ we apply (6.2) to obtain the sought result $\mathbf{R}_v(0_-)$. If the layer in $(0, d)$ is homogeneous then $\mathbf{R}_v' = 0$ in $(0, d)$ and hence $\mathbf{R}_v(0_+) = \mathbf{R}_v(d_-)$. In any case, the jump of $\mathbf{R}_v$ is given by (6.2) where, $\mathbf{H}_v^I, \ldots, \mathbf{H}_v^{IV}$ at z are given by

$$\mathbf{H}_v(z) = \operatorname{diag}\left[\mathbf{E}_v^-, \mathbf{E}_v^+\right]\mathbf{P}_v^{-1}(z_-)\mathbf{P}_v(z_+)\operatorname{diag}\left[\mathbf{E}_v^+, \mathbf{E}_v^-\right].$$

10. Comments and conclusions

The matrices Σ and $\mathbf{R}$ require a partition of the unknown, **s** or **v**, in forward- and backward-parts, a procedure which is usually referred to as wave splitting. A direct interpretation of the reflection process is then more immediate with Σ and $\mathbf{R}$. Also, the boundary values of Σ or $\mathbf{R}$ are necessary to determine the initial value for the Cauchy problem. Meanwhile, numerical purposes indicate that $\mathbf{Y}$ and **w** are more convenient in that the equation for $\mathbf{Y}$ does not involve derivatives of the material parameters. Also, it is an advantage of using the $\mathbf{Y}$ matrix that no wave splitting is required. This makes the approach more general in that wave splitting is not always allowed for dissipative solids. Further, working with $\mathbf{Y}$ avoids the evaluation of the eigenvectors $\mathbf{p}_\alpha$, within a layer, and is not affected by the occurrence of turning points.

In essence, three new results emerge from this paper. First, the connection between the $\mathbf{Y}, \Sigma, \mathbf{R}$ allows us to relate precisely the results obtained through different procedures. Second, account for both inhomogeneity and discontinuities provides a more general scheme for reflection-transmission in multilayers. Third, solving a Riccati equation for a reflection-transmission problem requires that we know the initial value which is in fact the value at the inhomogenity. This is performed by relating the boundary conditions (free surface, fixed surface, homogeneous half-space) to the appropriate value of $\mathbf{Y}, \Sigma, \mathbf{R}$.

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